In [1]:
```
i = 1
while i<=5:   # condition
    print('data science')
    i = i + 1  # increment
```

```
data science
data science
data science
data science
data science
```

In [2]:
```
i = 1

while i<=5:
    print('data science') # when we mention end then new line will not create
    j = 1
    while j<=4:
        print('technology')
        j = j + 1

    i = i + 1
    print()

    # the output which we got is very lengty but how to make them one line lets
```

```
data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology
```

In [3]:
```
i = 1
while i<=5:
    print(' datascience', end = "") # when we mention end then new line will not
```

```
    j = 1
    while j<=4:
        print(' technology', end="")
        j = j + 1

    i = i + 1
    print()
```

```
datascience technology technology technology technology
datascience technology technology technology technology
datascience technology technology technology technology
datascience technology technology technology technology
datascience technology technology technology technology
```

In [4]:
```
i = 1   # initializing

while i<=5:   # condition
    print('data science')
    i = i + 1 # increment
```

```
data science
data science
data science
data science
data science
```

In [5]:
```
i = 1   # initializing

while i<=10:   # condition
    print('data science',':', i)
    i = i + 1 # increment
```

```
data science : 1
data science : 2
data science : 3
data science : 4
data science : 5
data science : 6
data science : 7
data science : 8
data science : 9
data science : 10
```

In [8]:
```
i = 1   # initializing

while i>=1:   # condition
    print('data science',':', i)
    i = i - 1 # decrement
```

```
data science : 1
```

Nested While loop

In [10]:
```
# nested while loop

i = 1

while i<=5:
    print('data science') # when we mention end then new line will not create
    j = 1
```

```
    while j<=4:
        print('technology')
        j = j + 1

    i = i + 1
    print()

    # the output which we got is very lengty but how to make them one line lets
```

```
data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology
```

In [11]:

```
# nested while loop

i = 1

while i<=5:
    print('data science', end = ' *^ ') # when we mention end then new line will
    j = 1
    while j<=4:
        print('technology', end = ' *^ ')
        j = j + 1

    i = i + 1
    print()

    # the output which we got is very lengty but how to make them one line lets
```

```
data science *^ technology *^ technology *^ technology *^ technology *^
data science *^ technology *^ technology *^ technology *^ technology *^
data science *^ technology *^ technology *^ technology *^ technology *^
data science *^ technology *^ technology *^ technology *^ technology *^
data science *^ technology *^ technology *^ technology *^ technology *^
```

In [12]:
```
# lets use while loop usig some numbers

i = 1

while i <= 2 :
    j = 0
    while  j <= 2 :
        print(i*j, end=" ")
        j += 1
    print()
    i += 1
```

```
0 1 2
0 2 4
```

In [13]:
```
# lets use while loop usig some numbers

i = 1
while i <= 4 :
    j = 0
    while  j <= 3 :
        print(i*j, end=" ")
        j += 1
    print()
    i += 1
```

```
0 1 2 3
0 2 4 6
0 3 6 9
0 4 8 12
```

For Loop

In [14]:
```
name = 'nit'

for i in name:
    print(i)
```

```
n
i
t
```

In [15]:
```
name = [1,2,3]

for i in name:
    print(i)
```

```
1
2
3
```

In [16]:
```
name = (1,2,3)
```

```
for i in name:
    print(i)
```

```
1
2
3
```

In [17]:
```
range(5)
```

Out[17]:
```
range(0, 5)
```

In [18]:
```
for i in range(5):
    print(i)
```

```
0
1
2
3
4
```

In [19]:
```
for i in range(1,10,3):
    print(i)
```

```
1
4
7
```

In [20]:
```
# print the number which divisibe by 5
for i in range(1,51):
    if i%5==0:
        print(i)
```

```
5
10
15
20
25
30
35
40
45
50
```

. break

. continue

. pass

In [21]:
```
import keyword
keyword.kwlist
```

Out[21]:
```
['False',
 'None',
 'True',
 'and',
 'as',
 'assert',
 'async',
 'await',
 'break',
 'class',
 'continue',
 'def',
 'del',
 'elif',
 'else',
 'except',
 'finally',
 'for',
 'from',
 'global',
 'if',
 'import',
 'in',
 'is',
 'lambda',
 'nonlocal',
 'not',
 'or',
 'pass',
 'raise',
 'return',
 'try',
 'while',
 'with',
 'yield']
```

In [22]:
```
for i in range(1,11):
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

In [23]:
```
for i in range(1,11):
    if i == 3:
       break   # it will stop the loop
    print(i)
```

```
1
2
```

In [24]:
```
for i in range(1,11):
    if i == 3:
```

```
        continue   # it will skip the condition value and continue thw loop
    print(i)
```

```
1
2
4
5
6
7
8
9
10
```

In [25]:
```
for i in range(1,11):
```

```
  Cell In[25], line 1
    for i in range(1,11):
                         ^
SyntaxError: incomplete input
```

In [26]:
```
for i in range(1,11):
    pass   # it will pass the error or step
```

In [27]:
```
# write the code user ask chocklet from vendor machne write the basic code
x = int(input('how many choclets you want:?'))

i = 1
while i<=x:
    print('choclete')
    i+=1
```

```
choclete
choclete
choclete
choclete
choclete
choclete
choclete
choclete
choclete
```

In [28]:
```
ava = 5 # the machine has only 5 choclet

x = int(input('how many choclets you want:?'))

i = 1
while i<=x:
    print('choclete')
    i+=1
```

```
choclete
choclete
choclete
choclete
choclete
choclete
choclete
choclete
choclete
```

In [36]:
```
available_choclet = 5 # the machine has only 10 candis

x = int(input('How many choclets user want:?'))

i = 1
while i<=x:

    if i>available_choclet: # we stop the execution but which code execution not
        break # break is statement | means jump out of the loop
    print('choclet')
    i += 1

print('out of stock & bye for now')
```

```
choclet
choclet
choclet
choclet
choclet
out of stock & bye for now
```

In [37]:
```
available_choclet = 5 # the machine has only 10 candis

x = int(input('How many choclets you want:?'))

i = 1
while i<=x:

    if i>available_choclet: # we stop the execution but which code execution not
        print('out of stock')
        break # break is statement | means jump out of the loop
    print('choclet')
    i += 1

print('bye for now')
```

```
choclet
choclet
choclet
choclet
choclet
out of stock
bye for now
```

you need to print the number from 1 to 50 but dont print the number which is divisible by 3 or 5

In [39]:
```
for i in range(1,51):
    if i%3==0:
        print(i)
print('end')
```

```
3
6
9
12
15
18
21
24
27
30
33
36
39
42
45
48
end
```

In [40]:
```python
for i in range(1,51):

    if i%3 == 0:
        continue
    print(i)
print('end')
```

```
1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29
31
32
34
35
37
38
40
41
43
44
46
47
49
50
end
```

In [41]:
```
for i in range(1,51):

    if i%3 == 0 or i%5 == 0:
        continue
    print(i)
#print('end')
# it will skip all the value which is divisible by 3 or 5
```

```
1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49
```

In [ ]: